

# Sumaiya Sinha

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Dhaka, Bangladesh · Open to remote & relocation

## PROFILE

ML Engineer with production experience in computer vision, multimodal AI, and NLP. Research background through UIUC++ (2024, 2026) in LLM evaluation, code translation, and program analysis. Strong track record of shipping end-to-end systems — from model training and fine-tuning to inference optimization and automation infrastructure.

## PROFESSIONAL EXPERIENCE

### Machine Learning Engineer — Bengal Byte (Remote)

Sep 2024–Present

Computer vision, multimodal AI, and ML infrastructure

#### LLM Judge — Evaluation Framework for CV/VLM Systems

- Built an automated LLM-as-a-judge evaluation framework for CV/VLM systems, measuring semantic consistency, visual grounding, and reasoning reliability beyond standard accuracy metrics.
- Designed controlled prompt-variation robustness tests; implemented multi-signal reliability scoring (semantic similarity, visual evidence verification, coherence checks) to surface hallucinations, bias, and unstable reasoning.
- Integrated HuggingFace Transformers for model loading and inference; designed evaluation to be model-agnostic across vision-language architectures.

#### German License Plate OCR System

- Evaluated SVTR-Base (STN-based rectification + custom augmentation), PP-OCRV5, and a from-scratch PyTorch CRNN to identify the best approach for German license plate OCR.
- Selected SVTR-Base as the production model after iterating across learning-rate schedules, two-stage fine-tuning, and ensemble methods; achieved 95.67% word accuracy / 98.29% character accuracy on the held-out test set.
- Built a CRNN baseline from scratch in PyTorch (91.10% exact match) to independently benchmark the fine-tuned transformer approach.
- Diagnosed validation/test accuracy divergence and engineered a custom augmentation pipeline (geometric distortion, deterioration, color jitter) tuned to maximize real-world test performance over validation metrics.

#### ML Training Infrastructure Automation (YOLOv11 Pipeline)

- Designed and built an end-to-end automation pipeline keeping GPU training servers continuously utilized, automating dataset retrieval, job scheduling, and result collection with no manual intervention.
- Engineered an automated job queue that parses hyperparameter rows, deduplicates completed runs, and launches new training jobs on idle servers; paired with server-monitoring automation tracking training/idle state.
- Built a results pipeline via Make.com webhook to auto-fetch training outputs to AWS S3; added post-training export converting weights from .pt to ONNX for TensorRT deployment.

## EDUCATION

### BRAC University, Dhaka, Bangladesh

May 2024

B.Sc. in Computer Science and Engineering (CSE)

CGPA: 3.81/4.00

Honors: Highest Distinction; Dean's List (3×); Vice Chancellor's List (3×) IELTS: 7.0

Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Algorithms, Linear Algebra, Probability & Statistics

## RESEARCH EXPERIENCE

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### UIUC++ Summer Research in Software Engineering (SRSE)

Jun 2026–Present

Supervisor: [Prof. Darko Marinov](#), [Wenxi Wang](#), [Lize Shao](#)

Remote

- Working towards a survey research on autoregressive and diffusion code generation models.

### UIUC++ Summer Research in Software Engineering (SRSE)

Apr–Aug 2024

Supervisors: [Prof. Darko Marinov](#), [Reyhan Jabbarvand](#)

Remote

- Worked on the Lost in Translation replication study; implemented LLM-generated Rust translation experiments and evaluation pipelines in Python.
- Compared translation outputs of GPT-based models with C2Rust and transpilation benchmarks to evaluate syntactic correctness and code fidelity.
- Proposed readability and trust metrics for LLM-generated code using Rust compiler docs and linter diagnostics.
- Labeled 30+ mutants in an open dataset for mutation-based evaluation.

### Thesis, BRAC University

2023-2024

Supervisor: [Dr. Farig Yousuf Sadeque](#)

S. Sinha et al., “Identifying Code-Mixed and Code-Switched Hateful Remarks on Social Media using NLP,” BRAC University, 2024.

- Proposed a multi-model Transformer framework integrating BERT, BanglaBERT, mBERT, and LSTM-GRU hybrids for detection of implicit and explicit sexist remarks in multilingual text.
- Designed a data preprocessing pipeline for Bangla-English code-mixing: token normalization, transliteration, and class rebalancing with focal loss.
- Developed a fine-tuning architecture in PyTorch with dynamic learning rate scheduling, dropout regularization, and weighted cross-entropy loss.
- Built a dataset of 7,000 annotated samples in collaboration with Bengali.ai; improved baseline F1-score by 12% and achieved 90% accuracy.
- Conducted ablation studies on contextual embeddings, showing BanglaBERT + LSTM fusion yields optimal performance for low-resource multilingual classification.

## TEACHING EXPERIENCE

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### Teaching Assistant — BRAC University

Oct 2022–Jan 2024

- Assisted in undergraduate ML and CS courses; graded projects and led tutorial sessions for 100+ students.
- Designed problem sets and held weekly consultations; received commendation from course instructor.

## TECHNICAL SKILLS

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**Languages:** Python, JavaScript.

**ML Frameworks:** PyTorch, TensorFlow, Keras, scikit-learn, HuggingFace Transformers, PaddleOCR

**Techniques:** Transfer Learning, Fine-Tuning, CNN, RNN, LSTM, GRU, YOLO, OCR, Cross-Lingual Modeling, LLM Evaluation, VLM, Prompt Engineering

**Inference & Deploy:** ONNX, TensorRT, Docker, AWS S3, Vast.ai

**Tools & Infra:** Git, Linux, VS Code, Roboflow, Anaconda, Make/Integromat, Google Sheets API